

Pablo Leyva

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EDUCATION

New Jersey Institute of Technology

B.S. Applied Mathematics & Computer Science

Newark, NJ

Expected: May 2027

Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Machine Learning, Neurosymbolic AI, Explainable AI, Deep Learning, Reinforcement Learning

EXPERIENCE

Research Engineer Intern, Mechanistic Interpretability Lab

Sep 2026 – Present | Newark, NJ

- Working with Kieran Murphy to build a **PyTorch** pipeline classifying all **144 GPT-2 attention heads** by cross-checking in **TransformerLens** and **Hugging Face** to quantify no-op attention behavior; 74% default to the BOS sink.
- Re-ran the 3,000 sequences of rare, common and byte-free tokens; 15 heads shifted oppositely in early and late layers.
- Running **causal experiments** of 1.2M sequences to isolate attention shift.

AI/ML Intern, Apple

May 2026 – Aug 2026 | Cupertino, CA

- Designed agentic-commerce use cases for Siri, using Apple Wallet and Event Passes data as context to anticipate user needs.
- Built a **multimodal synthesizer** fusing speech and **On-Screen Awareness** into instruction prompts.
- Proposed a screenless human-computer interface that delegates work to recursive subagents on the user's behalf.

Machine Learning Intern, Angeles Investors

Jan 2026 – May 2026 | New York, NY

- Deployed an **asynchronous FastAPI** data ingestion pipeline via **Docker** that **parallelized research agents** and streamed structured results in real time, accelerating startup screening and enrichment.
- Productized AI startup diligence **agent swarm** powered by GPT-4o which led to a **\$500K pre-seed deal** at a \$48M pre-money valuation.

President, Society of Hispanic Professional Engineers (SHPE)

Sep 2025 – May 2026 | Newark, NJ

- Led 13 officers in raising **\$40,000** across **90+** events year-round, growing membership to a record **250+**.
- Architected and shipped the chapter's **first React Native app** in **27 days** while leading 3 engineers, scaling to **200+ users** in its **first month** with event registration, member details, and mentorship-program features.

AI/ML Intern, Apple

May 2025 – Aug 2025 | Cupertino, CA

- Deployed a **LangGraph** agent integrating **RAG** with **MCP** to reason over technical docs for Apple Pay merchants.
- Used **Qdrant** for **multilingual similarity search** and **GraphQL** to fetch and manipulate data across internal systems.
- Built an **LLM evaluation pipeline** benchmarking agent trajectories, tool-use success, groundedness, and hallucinations.

Software Engineering Intern, Caterpillar Inc.

May 2024 – Aug 2024 | Chicago, IL

- Implemented a complete **ETL workflow** in **Apache Spark** for Autonomous Vehicle Fleet Management, handling data validation, transformation, and storage in **PostgreSQL** with a **CI/CD** pipeline.
- Served end-to-end data pipelines via **FastAPI** microservices at **sub-100ms** latency for 10K+ daily requests.

PROJECTS

Evolv | Devpost

Flask, NumPy, Pandas, Python

- Placed **1st of 546** teams at HackPrinceton, building an entire backend and predictive modeling system in under 12 hours.
- Stochastically modeled** GPU compute demand and power consumption to forecast longterm regional impact.
- Productionized a **real-time Kubernetes pipeline** with LLM-generated reports, end-to-end running under 5 seconds.

Founder – HAI

Transformers, PyTorch, pgvector, Redis, Docker, FastAPI, LangGraph, Python

- Commercialized an AI-powered SAT prep platform, selling to NJ tutoring centers supporting **600 students** concurrently.
- Built **multimodal** input/output pipelines for PDFs, images, and text, combining **OCR**, vector search, semantic retrieval, prompt orchestration, and **GPT-4o API** to produce personalized questions, explanations, and study material.
- Implemented a **symbolic verifier** for mathematical reasoning on CoT yielding **100%** solution correctness before delivery.

Criteo Uplift: Large-Scale Causal ML | [GitHub](#) | [Kaggle](#)

Scikit-learn, XGBoost, EconML, SHAP, Python

- Trained Logistic Regression, Random Forest, and **gradient-boosted classifiers** across nearly **14M** advertising records, handling an extreme 0.29% conversion rate with class-aware preprocessing and model selection.
- Evaluated model generalization and achieved the following: 5-fold cross-validation ROC-AUC of **0.791 ± 0.003**, PR-AUC of **0.064** and Brier score of **0.0027** on held-out conversion prediction.

SKILLS

Languages: Python, SQL, C++, JavaScript/TypeScript, Swift

AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, LangGraph, RAG, MCP, Qdrant

Systems & Infrastructure: Docker, Git, Kubernetes, PostgreSQL, REST, GraphQL, FastAPI, Apache Spark, CI/CD, NumPy, Pandas